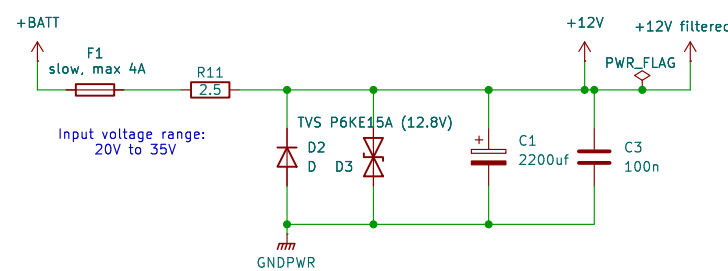
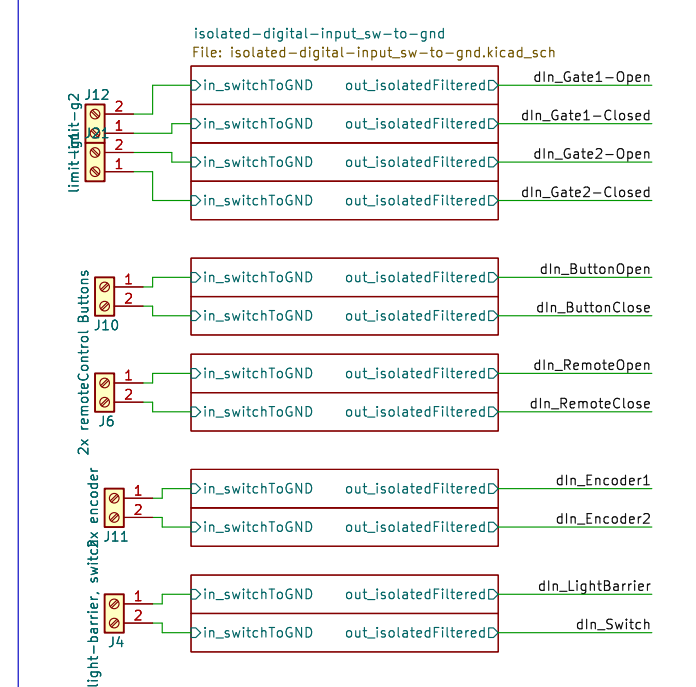


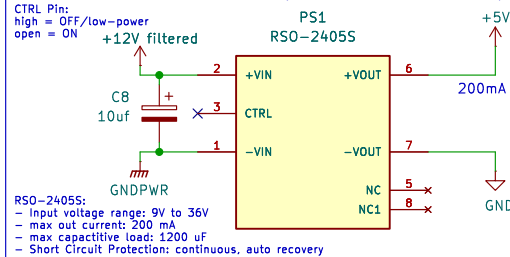
## 12V Supply (voltage filter)



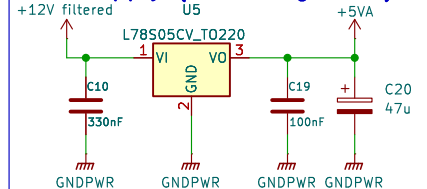
## Digital inputs (switch to GND, 12V pullup)



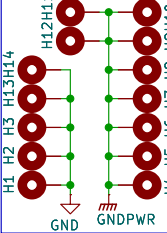
## 5V Isolated Supply (dc-dc converter)



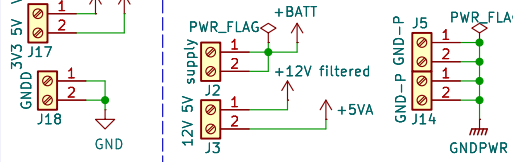
## 5V Supply (linear regulator)



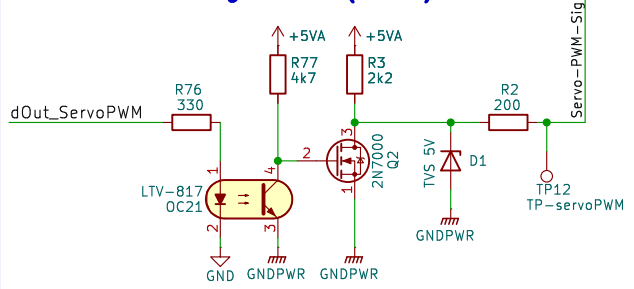
## Mountholes



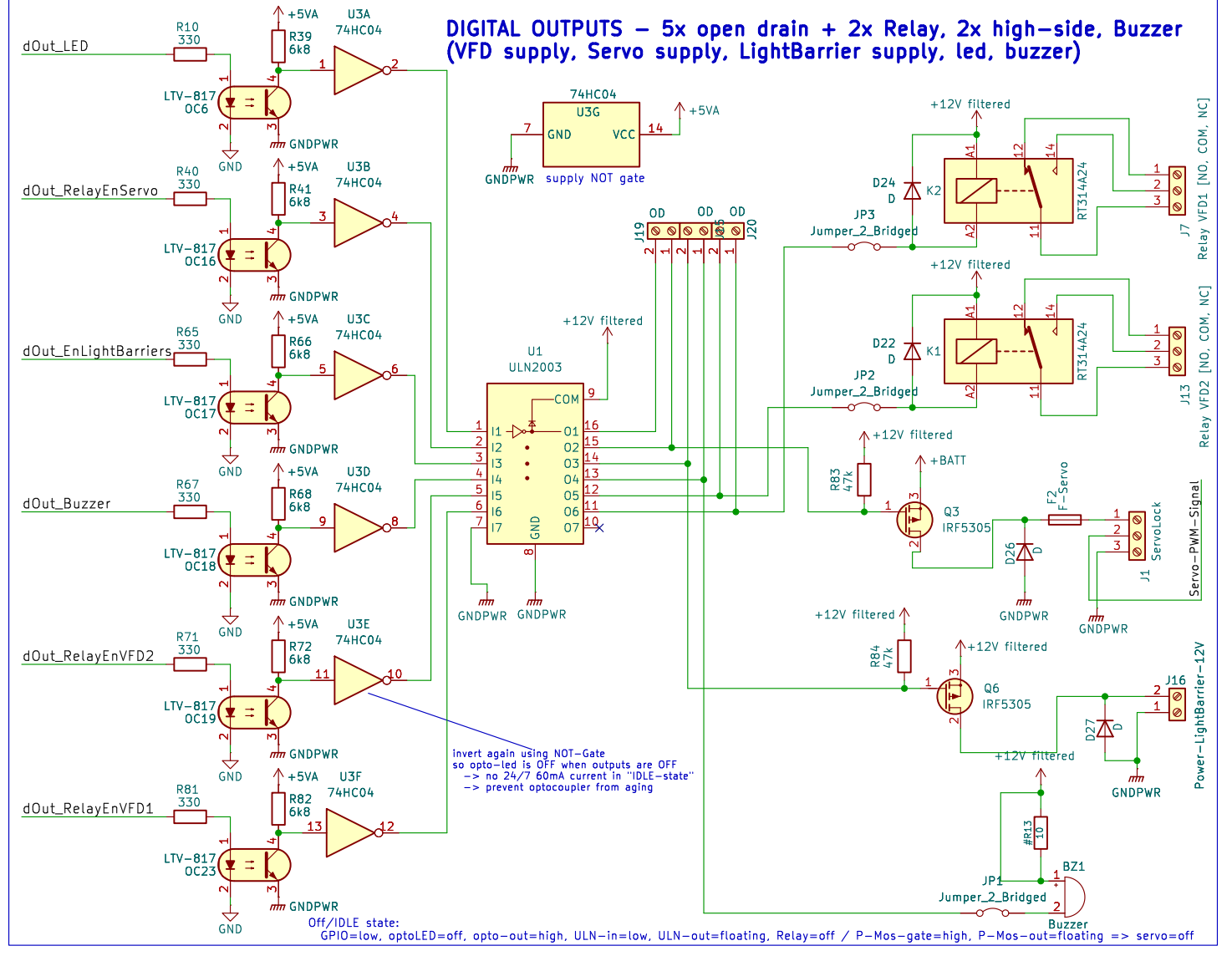
## Power terminals



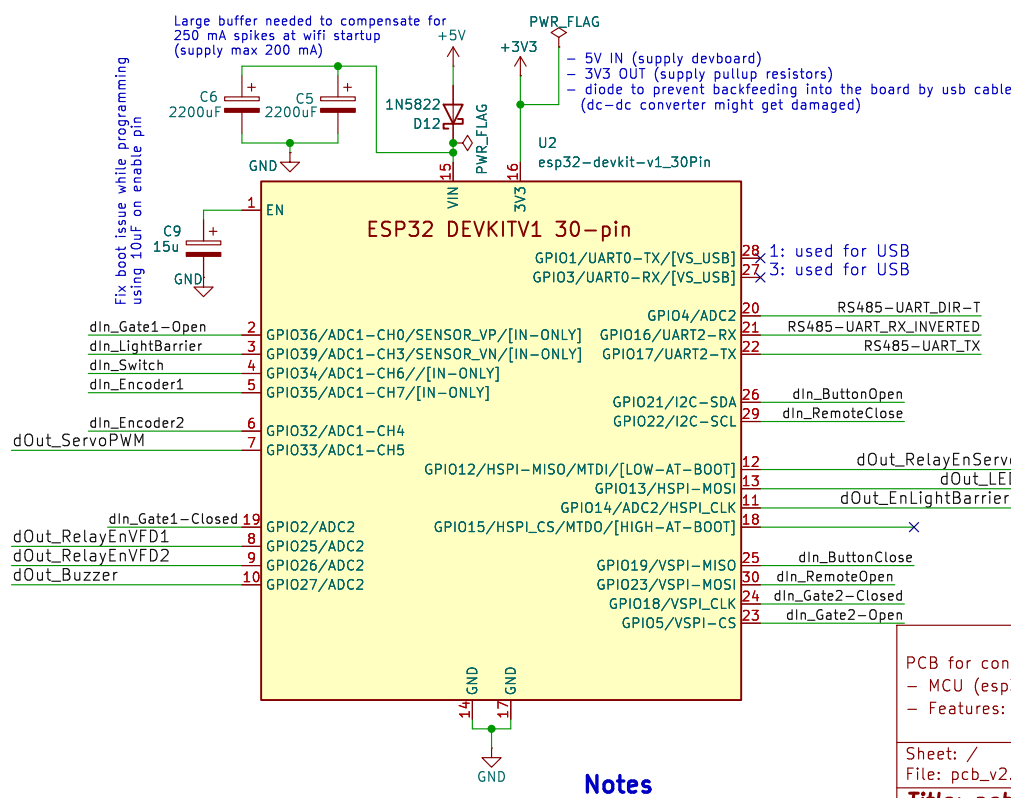
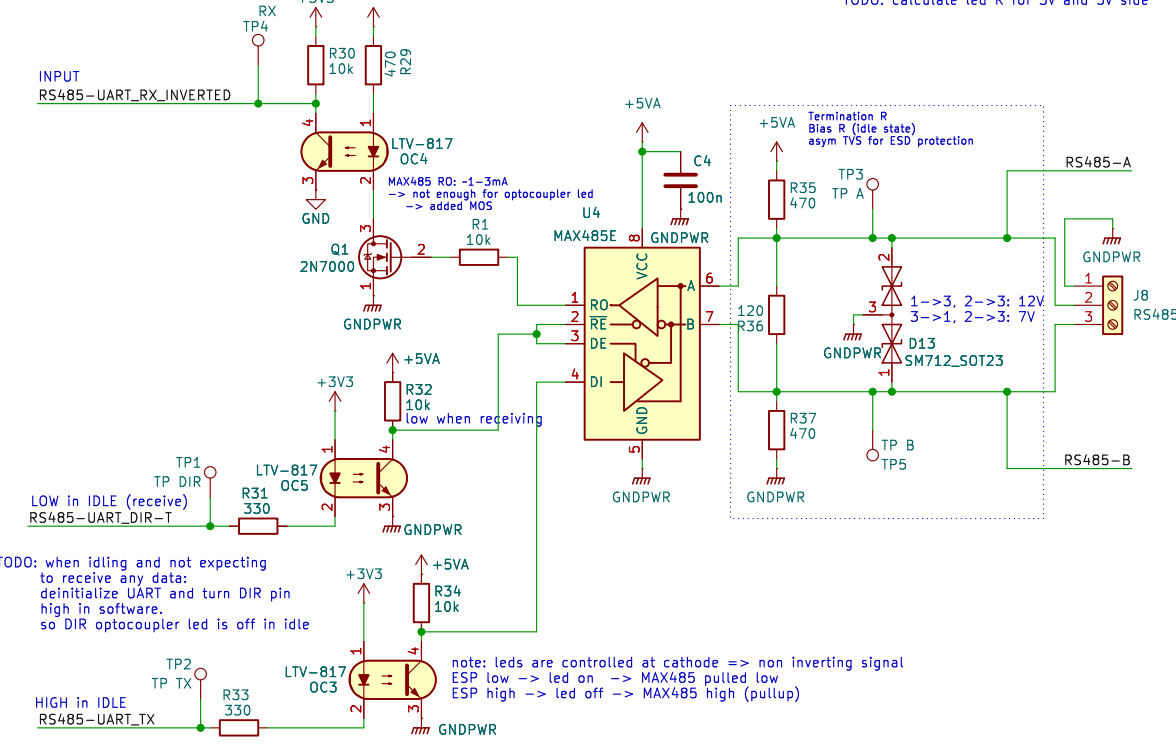
## Isolated PWM Signal out (Servo)



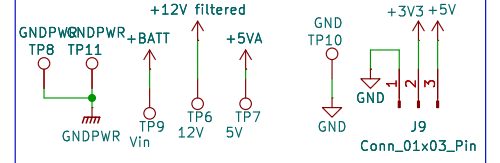
## DIGITAL OUTPUTS – 5x open drain + 2x Relay, 2x high-side, Buzzer (VFD supply, Servo supply, LightBarrier supply, led, buzzer)



## UART <=> RS485



## Testpoints power



## TODO

- [x] test if light barriers pull down when obstructed or free  
-> DONE: they pull to GND when obstructed -> no circuit necessary, connect both in parallel
- [x] measure light barrier standby current, if high add circuit to enable only when closing  
-> DONE: Photoelectric switch 12-24V NPN version <https://www.aliexpress.com/item/1005001615723145.html>
- [x] test esp32 board with isolated supply + diode  
- when starting wifi at boot draws ~250 mA power insufficient -> boot loop  
- when adding 2 x 2200 uF C after diode to buffer the peak it works reliably  
- when running with wifi on draws ~125 mA continuous  
- => add additional C (done) + prepare footprint for second DC DC converter in parallel?
- [x] test digital input circuit (lowpass+optocoupler work?)  
- works well, lowpass turning off (open switch, pulls up to 12V) is faster than expected though due to RC charging up over led, see notes schematic sub-sheet
- [x] test / evaluate optocoupler LED R with 3V3 => 330 Ohm -> 5 mA

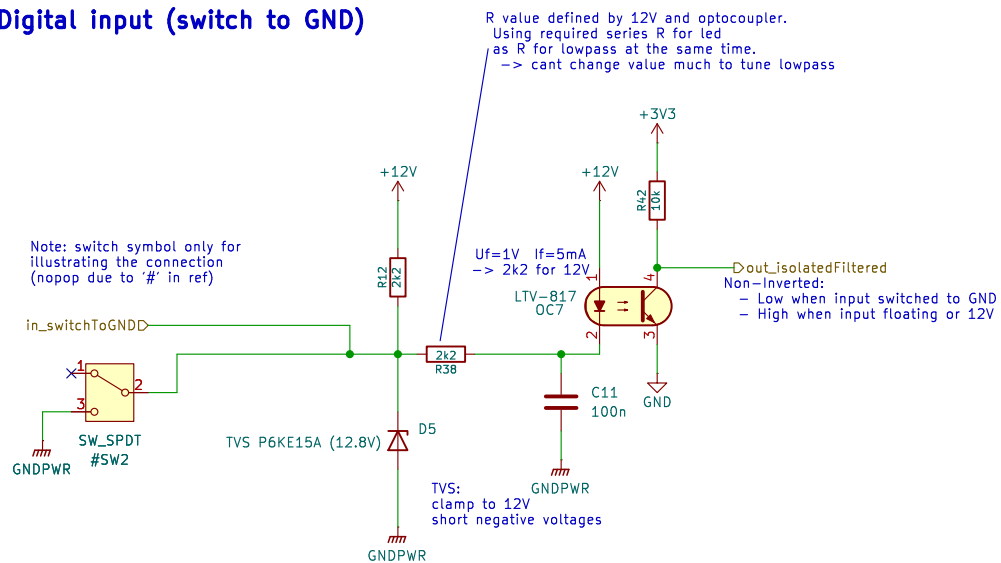
PCB for controlling a custom sliding gate:  
- MCU (esp32) fully isolated due to extreme EMI from near VFDs and long cables  
- Features: RS485, 12x digital-in, 6x openDrain-out (2x PMOS 12V, 2x Relay, 1x Buzzer)

|                                                |                  |          |
|------------------------------------------------|------------------|----------|
| Sheet: /                                       | Date: 2024-12-29 | Rev:     |
| File: pcb_v2.0-isolated-gate-control.kicad_sch |                  | Id: 1/13 |
| Size: A3                                       |                  |          |
| KiCad E.D.A. 8.0.7                             |                  |          |

## Notes

Measurements with Optocoupler and 24V relay connected (15mA)  
5 V, 200 Ohm -> Lled = 12 mA  
5 V, 510 Ohm -> Lled = 7 mA, Uce=0.7V  
5 V, 820 Ohm -> Lled = 4.5 mA, Uce=1.8V

## Digital input (switch to GND)



Notes made during testing this circuit:

- Lowpass is extremely asymmetric due to C additionally directly charging up over optocoupler led.  
-> better split R into two (one part between C and LED as well)?
- R\_filter = 3k3, C = 10uF:
  - 3.8ms delay turning on (switch closes to GND)
  - 0.1ms delay turning off (switch opens)
- R = 2k2, C = 10uF
  - 2.5ms delay turning on
  - 0.5ms delay turning off
- switch bouncing is filtered well when using >= 1uF

Sheet: /isolated-digital-input\_sw-to-gnd/  
File: isolated-digital-input\_sw-to-gnd.kicad\_sch

### Title:

Size: A4

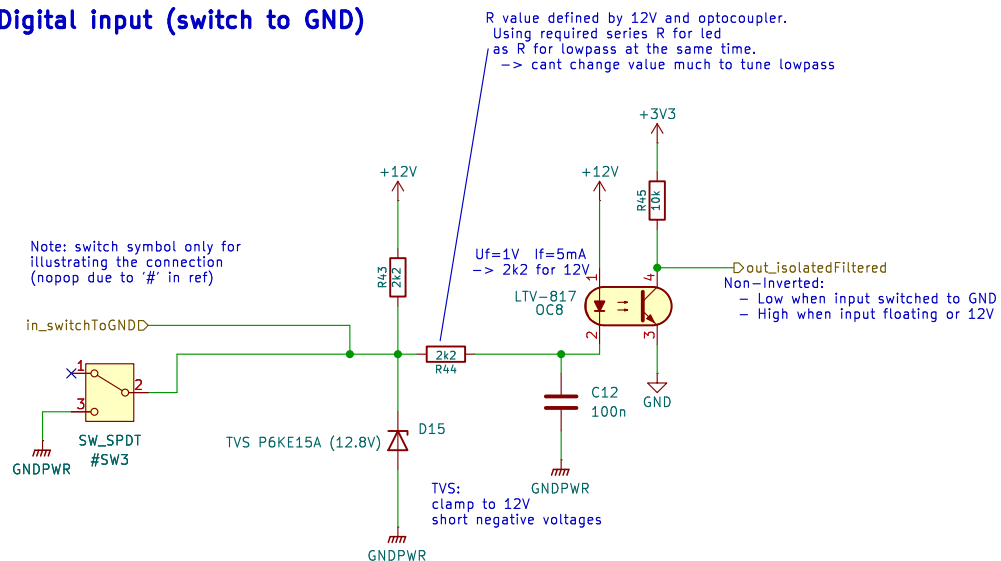
Date:

Rev:

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Id: 2/13

## Digital input (switch to GND)



Notes made during testing this circuit:

- Lowpass is extremely asymmetric due to C additionally directly charging up over optocoupler led.  
-> better split R into two (one part between C and LED as well)?
- R\_filter = 3k3, C = 10uF:
  - 3.8ms delay turning on (switch closes to GND)
  - 0.1ms delay turning off (switch opens)
- R = 2k2, C = 10uF
  - 2.5ms delay turning on
  - 0.5ms delay turning off
- switch bouncing is filtered well when using >= 1uF

Sheet: /isolated-digital-input\_sw-to-gnd1/  
File: isolated-digital-input\_sw-to-gnd.kicad\_sch

**Title:**

Size: A4

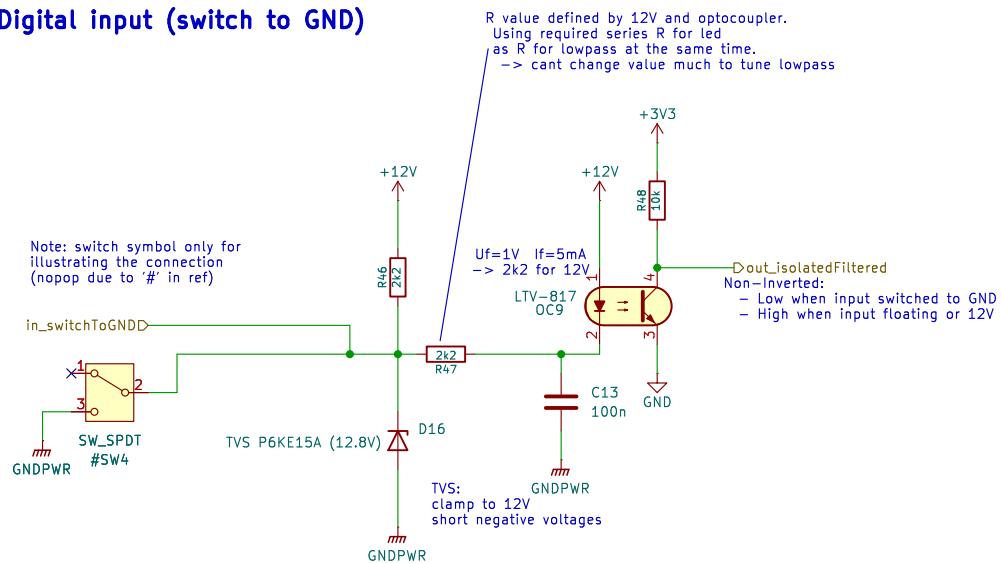
Date:

**Rev:**

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Id: 3/13

## Digital input (switch to GND)



Notes made during testing this circuit:

- Lowpass is extremely asymmetric due to C additionally directly charging up over optocoupler led.  
-> better split R into two (one part between C and LED as well)?
- R\_filter = 3k3, C = 10uF:
  - 3.8ms delay turning on (switch closes to GND)
  - 0.1ms delay turning off (switch opens)
- R = 2k2, C = 10uF
  - 2.5ms delay turning on
  - 0.5ms delay turning off
- switch bouncing is filtered well when using >= 1uF

Sheet: /isolated-digital-input\_sw-to-gnd2/  
File: isolated-digital-input\_sw-to-gnd.kicad\_sch

### Title:

Size: A4

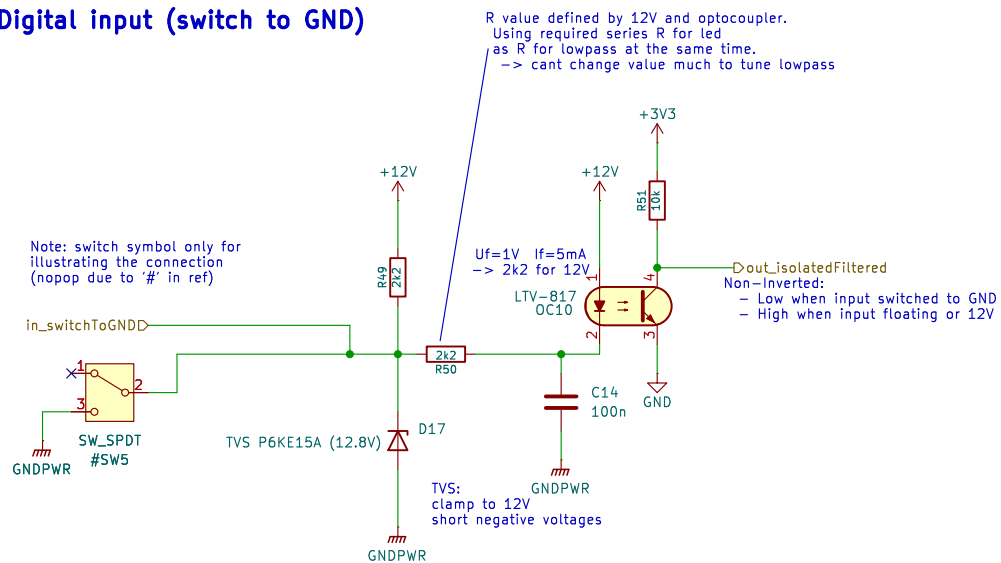
Date:

Rev:

KiCad E.D.A. 8.0.7

Id: 4/13

## Digital input (switch to GND)



Notes made during testing this circuit:

- Lowpass is extremely asymmetric due to C additionally directly charging up over optocoupler led.  
-> better split R into two (one part between C and LED as well)?
- R\_filter = 3k3, C = 10uF:
  - 3.8ms delay turning on (switch closes to GND)
  - 0.1ms delay turning off (switch opens)
- R = 2k2, C = 10uF
  - 2.5ms delay turning on
  - 0.5ms delay turning off
- switch bouncing is filtered well when using >= 1uF

Sheet: /isolated-digital-input\_sw-to-gnd3/  
File: isolated-digital-input\_sw-to-gnd.kicad\_sch

### Title:

Size: A4

Date:

Rev:

KiCad E.D.A. 8.0.7

Id: 5/13

### Digital input (switch to GND)

R value defined by 12V and optocoupler.  
Using required series R for led  
as R for lowpass at the same time.  
-> cant change value much to tune lowpass

Note: switch symbol only for illustrating the connection (nopop due to '#' in ref)

in\_switchToGND

SW\_SPDT #SW6

GNDPWR

TVS P6KE15A (12.8V)

D18

GNDPWR

R52 2k2

2k2 R53

LTV-817 OC11

Uf=1V If=5mA  
-> 2k2 for 12V

C15 100n

GNDPWR

+12V

+12V

+3V3

R54 10k

Out\_IsolatedFiltered

Non-Inverted:  
- Low when input switched to GND  
- High when input floating or 12V

Notes made during testing this circuit:

- Lowpass is extremely asymmetric due to C additionally directly charging up over optocoupler led.  
-> better split R into two (one part between C and LED as well)?
- R\_filter = 3k3, C = 10uF:
  - 3.8ms delay turning on (switch closes to GND)
  - 0.1ms delay turning off (switch opens)
- R = 2k2, C = 10uF
  - 2.5ms delay turning on
  - 0.5ms delay turning off
- switch bouncing is filtered well when using >= 1uF

in\_switchToGNDD

SW\_SPDT  
#CWF

TVS P6KE15A (12.8V)

0.8V) 

7

TVS: clamp to 12V  
short negative voltages

R value defined by 12V and optocoupler.  
Using required series R for led  
as R for lowpass at the same time.  
-> cant change value much to tune lowpass

$U_f = 1V$   $I_f = 5mA$   
 $\rightarrow 2k\Omega$  for  $12V$

LIV=817  
OC11

$C_1$



+3V



7

→ out\_isolatedFiltered

Non-Inverted:

- Low when input switched to GND
- High when input floating or 12V

Notes made during testing this circuit:

- Lowpass is extremely asymmetric due to C additionally directly charging up over optocoupler led.  
-> better split R into two (one part between C and LED as well)?
- R\_filter = 3k $\Omega$ , C = 10 $\mu$ F:
  - 3.8ms delay turning on (switch closes to GND)
  - 0.1ms delay turning off (switch opens)
- R = 2k $\Omega$ , C = 10 $\mu$ F
  - 2.5ms delay turning on
  - 0.5ms delay turning off
- switch bouncing is filtered well when using  $\geq 1\mu$ F

Sheet: /isolated-digital-input\_sw-to-gnd4/  
File: isolated-digital-input\_sw-to-gnd.kicad\_sch

**Title:**

Size: A4

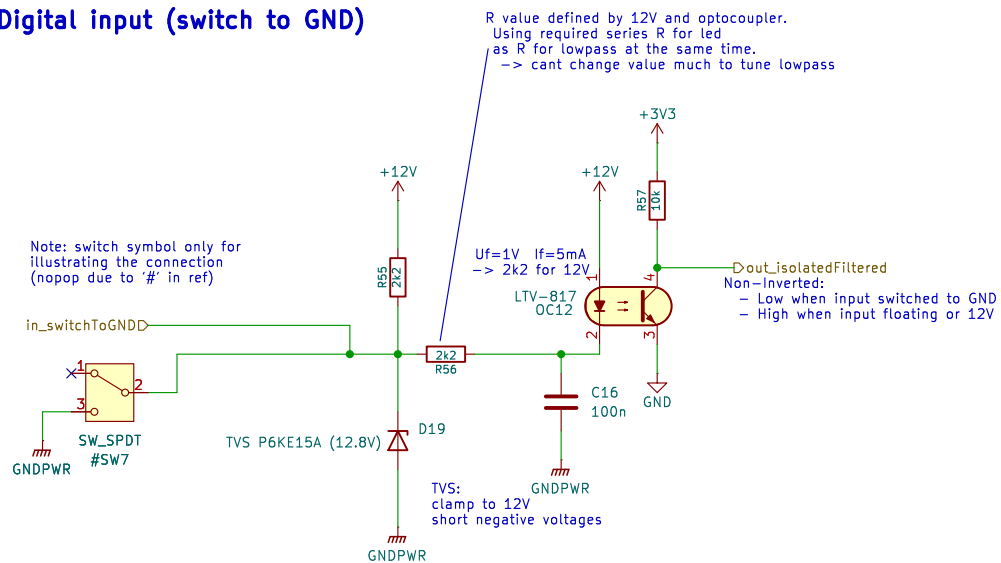
Date:

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Rev:

Id: 6/13

## Digital input (switch to GND)



Notes made during testing this circuit:

- Lowpass is extremely asymmetric due to C additionally directly charging up over optocoupler led.  
-> better split R into two (one part between C and LED as well)?
- R\_filter = 3k3, C = 10uF:
  - 3.8ms delay turning on (switch closes to GND)
  - 0.1ms delay turning off (switch opens)
- R = 2k2, C = 10uF
  - 2.5ms delay turning on
  - 0.5ms delay turning off
- switch bouncing is filtered well when using >= 1uF

Sheet: /isolated-digital-input\_sw-to-gnd5/  
File: isolated-digital-input\_sw-to-gnd.kicad\_sch

**Title:**

Size: A4

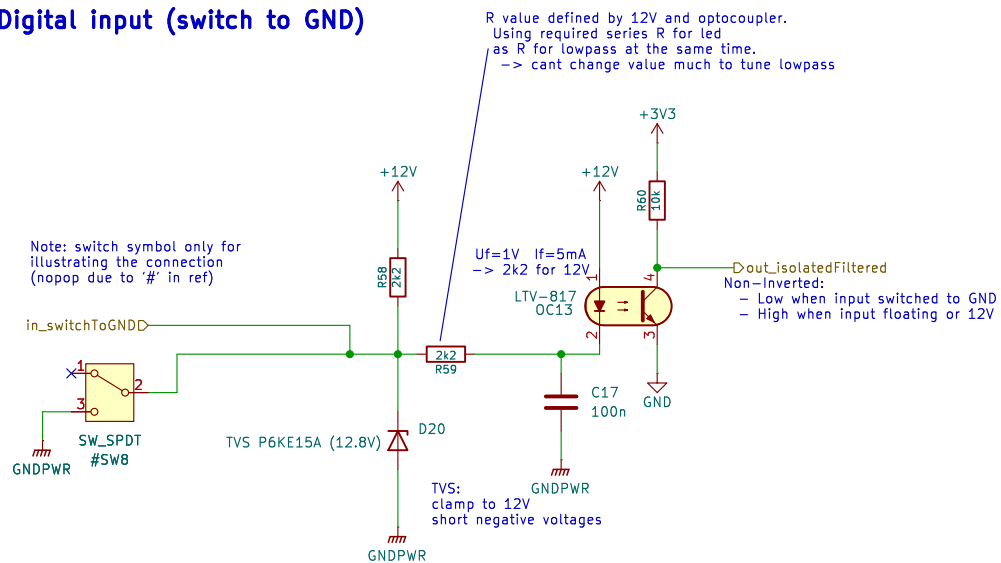
Date:

**Rev:**

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Id: 7/13

## Digital input (switch to GND)



Notes made during testing this circuit:

- Lowpass is extremely asymmetric due to C additionally directly charging up over optocoupler led.  
-> better split R into two (one part between C and LED as well)?
- R\_filter = 3k3, C = 10uF:
  - 3.8ms delay turning on (switch closes to GND)
  - 0.1ms delay turning off (switch opens)
- R = 2k2, C = 10uF
  - 2.5ms delay turning on
  - 0.5ms delay turning off
- switch bouncing is filtered well when using >= 1uF

Sheet: /isolated-digital-input\_sw-to-gnd6/  
File: isolated-digital-input\_sw-to-gnd.kicad\_sch

### Title:

Size: A4

Date:

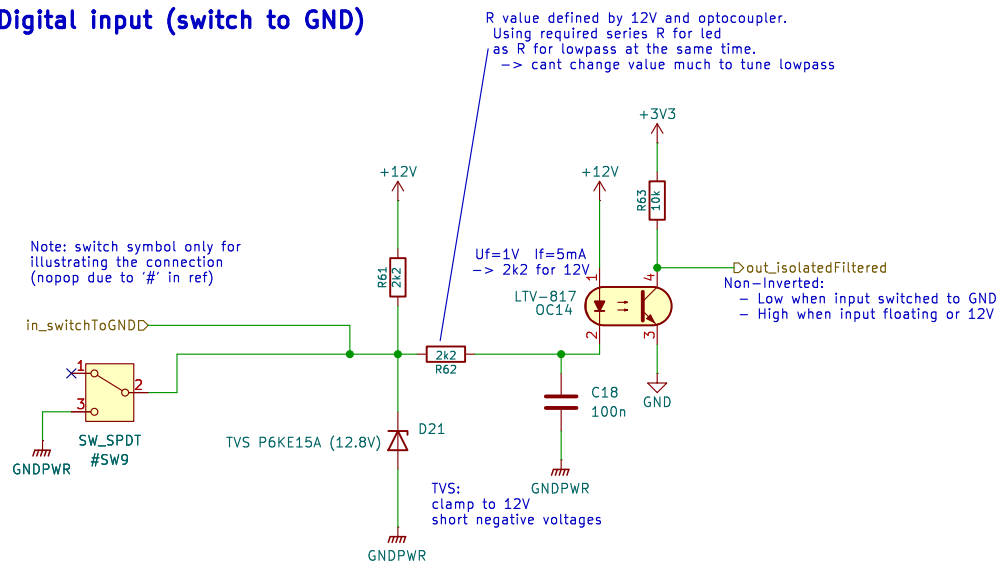
Rev:

KiCad E.D.A. 8.0.7

Id: 8/13



## Digital input (switch to GND)



Notes made during testing this circuit:

- Lowpass is extremely asymmetric due to C additionally directly charging up over optocoupler led.  
-> better split R into two (one part between C and LED as well)?
- R\_filter = 3k3, C = 10uF:
  - 3.8ms delay turning on (switch closes to GND)
  - 0.1ms delay turning off (switch opens)
- R = 2k2, C = 10uF
  - 2.5ms delay turning on
  - 0.5ms delay turning off
- switch bouncing is filtered well when using >= 1uF

Sheet: /isolated-digital-input\_sw-to-gnd7/  
File: isolated-digital-input\_sw-to-gnd.kicad\_sch

### Title:

Size: A4

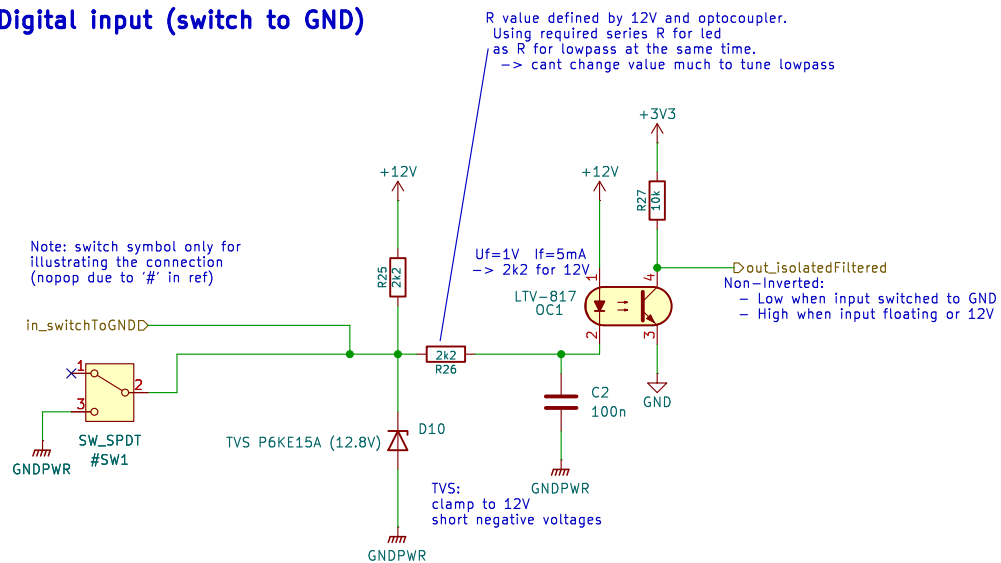
Date:

Rev:

KiCad E.D.A. 8.0.7

Id: 9/13

## Digital input (switch to GND)



Notes made during testing this circuit:

- Lowpass is extremely asymmetric due to C additionally directly charging up over optocoupler led.  
-> better split R into two (one part between C and LED as well)?
- R\_filter = 3k3, C = 10uF:
  - 3.8ms delay turning on (switch closes to GND)
  - 0.1ms delay turning off (switch opens)
- R = 2k2, C = 10uF
  - 2.5ms delay turning on
  - 0.5ms delay turning off
- switch bouncing is filtered well when using >= 1uF

Sheet: /isolated-digital-input\_sw-to-gnd8/  
File: isolated-digital-input\_sw-to-gnd.kicad\_sch

### Title:

Size: A4

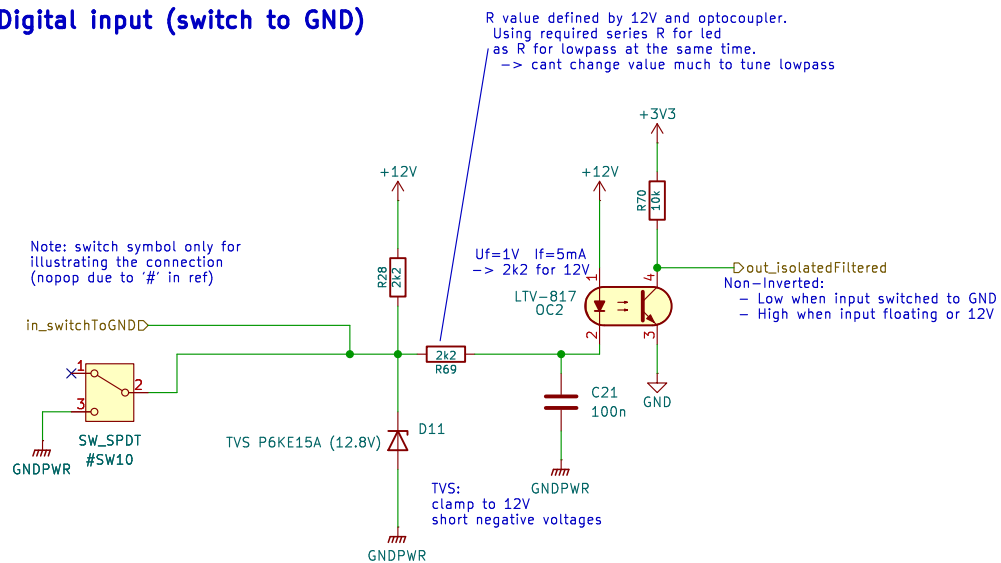
Date:

Rev:

KiCad E.D.A. 8.0.7

Id: 10/13

## Digital input (switch to GND)



Notes made during testing this circuit:

- Lowpass is extremely asymmetric due to C additionally directly charging up over optocoupler led.  
-> better split R into two (one part between C and LED as well)?
- R\_filter = 3k3, C = 10uF:
  - 3.8ms delay turning on (switch closes to GND)
  - 0.1ms delay turning off (switch opens)
- R = 2k2, C = 10uF
  - 2.5ms delay turning on
  - 0.5ms delay turning off
- switch bouncing is filtered well when using >= 1uF

Sheet: /isolated-digital-input\_sw-to-gnd9/  
File: isolated-digital-input\_sw-to-gnd.kicad\_sch

### Title:

Size: A4

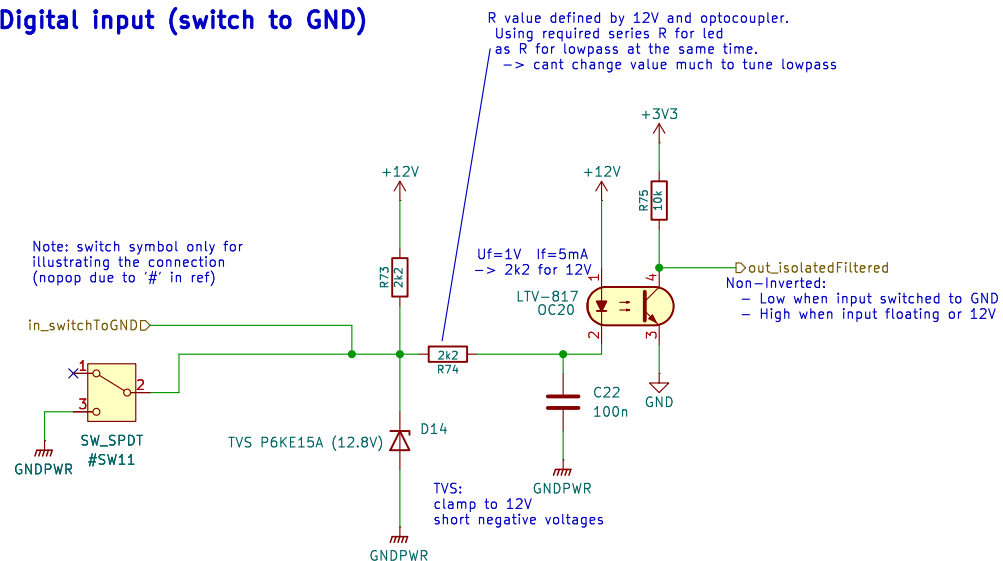
Date:

Rev:

KiCad E.D.A. 8.0.7

Id: 11/13

## Digital input (switch to GND)



Notes made during testing this circuit:

- Lowpass is extremely asymmetric due to C additionally directly charging up over optocoupler led.  
-> better split R into two (one part between C and LED as well)?
- R\_filter = 3k3, C = 10uF:
  - 3.8ms delay turning on (switch closes to GND)
  - 0.1ms delay turning off (switch opens)
- R = 2k2, C = 10uF
  - 2.5ms delay turning on
  - 0.5ms delay turning off
- switch bouncing is filtered well when using >= 1uF

Sheet: /isolated-digital-input\_sw-to-gnd10/  
File: isolated-digital-input\_sw-to-gnd.kicad\_sch

### Title:

Size: A4

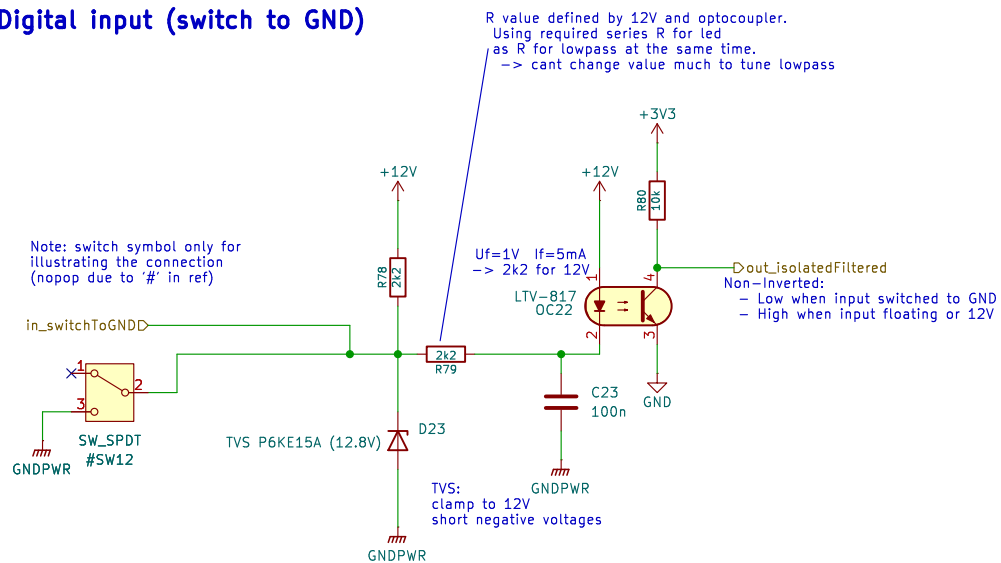
Date:

Rev:

KiCad E.D.A. 8.0.7

Id: 12/13

## Digital input (switch to GND)



Notes made during testing this circuit:

- Lowpass is extremely asymmetric due to C additionally directly charging up over optocoupler led.  
-> better split R into two (one part between C and LED as well)?
- R\_filter = 3k3, C = 10uF:
  - 3.8ms delay turning on (switch closes to GND)
  - 0.1ms delay turning off (switch opens)
- R = 2k2, C = 10uF
  - 2.5ms delay turning on
  - 0.5ms delay turning off
- switch bouncing is filtered well when using >= 1uF

Sheet: /isolated-digital-input\_sw-to-gnd11/  
File: isolated-digital-input\_sw-to-gnd.kicad\_sch

### Title:

Size: A4

Date:

Rev:

KiCad E.D.A. 8.0.7

Id: 13/13